Facilities AI Field Lab Scenario Summary

Current labs, changing situations, and learner questions

This document summarizes the active training labs. Each lab uses a realistic facilities situation as context while the learner practices practical AI use: organizing requests, finding missing information, checking sources, drafting handoffs, protecting sensitive information, and keeping human review in place. Each lab has eight written-response questions across four changing situations.

Active Lab Overview

Coach safe AI use

Supervisors | Leadership | 8 questions

Intake and scope → Source review → Draft and critique → Handoff

Use AI with an air-handler manual

HVAC | Foundations | 8 questions

Intake and scope → Source review → Draft and critique → Handoff

Use AI with BAS trends

BAS / Controls | Intermediate | 8 questions

Intake and scope → Source review → Draft and critique → Handoff

Use AI for electrical triage

Electrical | Safety-critical | 8 questions

Intake and scope → Source review → Draft and critique → Handoff

Use AI for a leak handoff

Plumbing | Foundations | 8 questions

Intake and scope → Source review → Draft and critique → Handoff

Use AI around life-safety impairments

Fire / Life Safety | Safety-critical | 8 questions

Intake and scope → Source review → Draft and critique → Handoff

Use AI to review roof notes

Building Envelope | Intermediate | 8 questions

Intake and scope → Source review → Draft and critique → Handoff

Use AI with an IAQ complaint

Custodial / IAQ | Foundations | 8 questions

Intake and scope → Source review → Draft and critique → Handoff

Use AI with irrigation data

Grounds / Irrigation | Foundations | 8 questions

Intake and scope → Source review → Draft and critique → Handoff

1. Coach safe AI use

Supervisors | Leadership | 8 questions

Practice reviewing a team member's AI-assisted work, giving useful feedback, and setting clear boundaries.

Scenario context

a technician's AI-assisted work order

Evidence source: ILLUSTRATIVE SUPERVISOR CASE · AI-ASSISTED WORK ORDER

Questions and changing situations

Question 1 | Intake & scope

Ask AI to clean up this request or note into a draft work order. Have it list the location, problem, facts, missing information, and next check.

> **Job situation:** A technician sends a draft work order after a warm-room complaint. It is polished, but the draft says "failed actuator" even though no actuator test was recorded.
>
> **What we know:** A technician used AI to summarize a complaint. The draft is fluent but includes one unsupported cause, omits a safety limitation, and does not cite the source note. The supervisor must coach the process without discouraging appropriate AI use.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source review and supervisor-approved next step.

Question 2 | Intake & scope

Ask AI what information is missing and have it write three short questions for the requester or technician.

> **Job situation:** A technician sends a draft work order after a warm-room complaint. It is polished, but the draft says "failed actuator" even though no actuator test was recorded.
>
> **What we know:** A technician used AI to summarize a complaint. The draft is fluent but includes one unsupported cause, omits a safety limitation, and does not cite the source note. The supervisor must coach the process without discouraging appropriate AI use.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source review and supervisor-approved next step.

Question 3 | Source review

Give AI the approved note, manual section, or procedure. Ask it to find the two or three useful points and give you the page or section number.

> **Job situation:** The supervisor receives the technician's field note, the AI draft, and a manufacturer sequence. The draft cites none of them.
>
> **What we know:** A technician used AI to summarize a complaint. The draft is fluent but includes one unsupported cause, omits a safety limitation, and does not cite the source note. The supervisor must coach the process without discouraging appropriate AI use.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source review and supervisor-approved next step.

Question 4 | Source review

AI made a guess about the cause. Ask: "What in the note supports that? What is missing? How should we word it if we are not sure?"

Job situation: The supervisor receives the technician's field note, the AI draft, and a manufacturer sequence. The draft cites none of them.

What we know: A technician used AI to summarize a complaint. The draft is fluent but includes one unsupported cause, omits a safety limitation, and does not cite the source note. The supervisor must coach the process without discouraging appropriate AI use.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source review and supervisor-approved next step.

Question 5 | Draft & critique

Ask AI to turn the checked facts into a short email, work-order update, or shift note. Tell it who will read it and what facts to include.

Job situation: The technician asks whether the AI draft can be copied directly into the maintenance system before the end of the shift.

What we know: A technician used AI to summarize a complaint. The draft is fluent but includes one unsupported cause, omits a safety limitation, and does not cite the source note. The supervisor must coach the process without discouraging appropriate AI use.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source review and supervisor-approved next step.

Question 6 | Draft & critique

Ask AI to sort the draft into three groups: what we saw, what we think, and what a qualified person still needs to approve.

Job situation: The technician asks whether the AI draft can be copied directly into the maintenance system before the end of the shift.

What we know: A technician used AI to summarize a complaint. The draft is fluent but includes one unsupported cause, omits a safety limitation, and does not cite the source note. The supervisor must coach the process without discouraging appropriate AI use.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source review and supervisor-approved next step.

Question 7 | Handoff

Ask AI to make a short review checklist for the supervisor or qualified person who must check the draft before it is used.

Job situation: A night-shift handoff must explain what was observed, what remains open, and who is responsible for the next check.

What we know: A technician used AI to summarize a complaint. The draft is fluent but includes one unsupported cause, omits a safety limitation, and does not cite the source note. The supervisor must coach the process without discouraging appropriate AI use.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source review and supervisor-approved next step.

Question 8 | Handoff

Ask AI to find and flag names, access details, security information, contract details, or other information that should not be shared.

Job situation: A night-shift handoff must explain what was observed, what remains open, and who is responsible for the next check.

What we know: A technician used AI to summarize a complaint. The draft is fluent but includes one unsupported cause, omits a safety limitation, and does not cite the source note. The supervisor must coach the process without discouraging appropriate AI use.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source review and supervisor-approved next step.

2. Use AI with an air-handler manual

HVAC | Foundations | 8 questions

Practice turning a manual and a startup note into a source-grounded checklist.

Scenario context

an AHU pre-start checklist

Evidence source: ILLUSTRATIVE SOURCE PACK · AHU START-UP GUIDE

Questions and changing situations

Question 1 | Intake & scope

Ask AI to clean up this request or note into a draft work order. Have it list the location, problem, facts, missing information, and next check.

> **Job situation:** A PM work order for AHU-3 says “unit ready for startup,” but the note has only one checkmark and no filter, damper, or condensate details.
>
> **What we know:** The manual lists filters, access panels, fan rotation, outdoor-air damper movement, and a primed condensate trap. It does not provide a universal discharge temperature.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source check and field observation.

Question 2 | Intake & scope

Ask AI what information is missing and have it write three short questions for the requester or technician.

> **Job situation:** A PM work order for AHU-3 says “unit ready for startup,” but the note has only one checkmark and no filter, damper, or condensate details.
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> **What we know:** The manual lists filters, access panels, fan rotation, outdoor-air damper movement, and a primed condensate trap. It does not provide a universal discharge temperature.
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> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source check and field observation.

Question 3 | Source review

Give AI the approved note, manual section, or procedure. Ask it to find the two or three useful points and give you the page or section number.

> **Job situation:** The technician has an approved startup checklist, a manufacturer IOM excerpt, and a photo of the unit nameplate with the serial number hidden.
>
> **What we know:** The manual lists filters, access panels, fan rotation, outdoor-air damper movement, and a primed condensate trap. It does not provide a universal discharge temperature.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source check and field observation.

Question 4 | Source review

AI made a guess about the cause. Ask: "What in the note supports that? What is missing? How should we word it if we are not sure?"

Job situation: The technician has an approved startup checklist, a manufacturer IOM excerpt, and a photo of the unit nameplate with the serial number hidden.

What we know: The manual lists filters, access panels, fan rotation, outdoor-air damper movement, and a primed condensate trap. It does not provide a universal discharge temperature.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source check and field observation.

Question 5 | Draft & critique

Ask AI to turn the checked facts into a short email, work-order update, or shift note. Tell it who will read it and what facts to include.

Job situation: AI drafts a startup checklist and adds a "normal discharge temperature" that does not appear in the supplied manual.

What we know: The manual lists filters, access panels, fan rotation, outdoor-air damper movement, and a primed condensate trap. It does not provide a universal discharge temperature.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source check and field observation.

Question 6 | Draft & critique

Ask AI to sort the draft into three groups: what we saw, what we think, and what a qualified person still needs to approve.

Job situation: AI drafts a startup checklist and adds a "normal discharge temperature" that does not appear in the supplied manual.

What we know: The manual lists filters, access panels, fan rotation, outdoor-air damper movement, and a primed condensate trap. It does not provide a universal discharge temperature.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source check and field observation.

Question 7 | Handoff

Ask AI to make a short review checklist for the supervisor or qualified person who must check the draft before it is used.

Job situation: The next technician needs a short handoff before energizing the unit on the morning shift.

What we know: The manual lists filters, access panels, fan rotation, outdoor-air damper movement, and a primed condensate trap. It does not provide a universal discharge temperature.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source check and field observation.

Question 8 | Handoff

Ask AI to find and flag names, access details, security information, contract details, or other information that should not be shared.

Job situation: The next technician needs a short handoff before energizing the unit on the morning shift.

What we know: The manual lists filters, access panels, fan rotation, outdoor-air damper movement, and a primed condensate trap. It does not provide a universal discharge temperature.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: source check and field observation.

3. Use AI with BAS trends

BAS / Controls | Intermediate | 8 questions

Practice asking AI to organize a comfort complaint without letting it diagnose from one trend.

Scenario context

a warm conference-room complaint

Evidence source: ILLUSTRATIVE CASE · CONFERENCE ROOM 204

Questions and changing situations

Question 1 | Intake & scope

Ask AI to clean up this request or note into a draft work order. Have it list the location, problem, facts, missing information, and next check.

> **Job situation:** A service request says Conference Room 204 is warm. The BAS trend shows a 76°F zone, 72°F setpoint, 55°F supply air, and 100% damper command.
>
> **What we know:** Zone temperature is 76°F, setpoint is 72°F, supply air is 55°F, and damper command is 100%. A command is not proof of physical airflow.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: actual airflow, physical position, and a reference reading.

Question 2 | Intake & scope

Ask AI what information is missing and have it write three short questions for the requester or technician.

> **Job situation:** A service request says Conference Room 204 is warm. The BAS trend shows a 76°F zone, 72°F setpoint, 55°F supply air, and 100% damper command.
>
> **What we know:** Zone temperature is 76°F, setpoint is 72°F, supply air is 55°F, and damper command is 100%. A command is not proof of physical airflow.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: actual airflow, physical position, and a reference reading.

Question 3 | Source review

Give AI the approved note, manual section, or procedure. Ask it to find the two or three useful points and give you the page or section number.

> **Job situation:** The analyst has a trend export, point list, sequence of operation, and a note that the damper command is not proof of physical airflow.
>
> **What we know:** Zone temperature is 76°F, setpoint is 72°F, supply air is 55°F, and damper command is 100%. A command is not proof of physical airflow.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: actual airflow, physical position, and a reference reading.

Question 4 | Source review

AI made a guess about the cause. Ask: "What in the note supports that? What is missing? How should we word it if we are not sure?"

Job situation: The analyst has a trend export, point list, sequence of operation, and a note that the damper command is not proof of physical airflow.

What we know: Zone temperature is 76°F, setpoint is 72°F, supply air is 55°F, and damper command is 100%. A command is not proof of physical airflow.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: actual airflow, physical position, and a reference reading.

Question 5 | Draft & critique

Ask AI to turn the checked facts into a short email, work-order update, or shift note. Tell it who will read it and what facts to include.

Job situation: AI writes "failed actuator" in the first sentence even though no airflow reading or physical inspection is attached.

What we know: Zone temperature is 76°F, setpoint is 72°F, supply air is 55°F, and damper command is 100%. A command is not proof of physical airflow.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: actual airflow, physical position, and a reference reading.

Question 6 | Draft & critique

Ask AI to sort the draft into three groups: what we saw, what we think, and what a qualified person still needs to approve.

Job situation: AI writes "failed actuator" in the first sentence even though no airflow reading or physical inspection is attached.

What we know: Zone temperature is 76°F, setpoint is 72°F, supply air is 55°F, and damper command is 100%. A command is not proof of physical airflow.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: actual airflow, physical position, and a reference reading.

Question 7 | Handoff

Ask AI to make a short review checklist for the supervisor or qualified person who must check the draft before it is used.

Job situation: The controls technician needs the next three checks in a clear handoff, not a replacement-parts recommendation.

What we know: Zone temperature is 76°F, setpoint is 72°F, supply air is 55°F, and damper command is 100%. A command is not proof of physical airflow.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: actual airflow, physical position, and a reference reading.

Question 8 | Handoff

Ask AI to find and flag names, access details, security information, contract details, or other information that should not be shared.

Job situation: The controls technician needs the next three checks in a clear handoff, not a replacement-parts recommendation.

What we know: Zone temperature is 76°F, setpoint is 72°F, supply air is 55°F, and damper command is 100%. A command is not proof of physical airflow.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: actual airflow, physical position, and a reference reading.

4. Use AI for electrical triage

Electrical | Safety-critical | 8 questions

Practice using AI to structure a safe report without asking it to replace electrical procedures.

Scenario context

a repeated breaker-trip report

Evidence source: ILLUSTRATIVE CASE · REPEATED BREAKER TRIP

Questions and changing situations

Question 1 | Intake & scope

Ask AI to clean up this request or note into a draft work order. Have it list the location, problem, facts, missing information, and next check.

> **Job situation:** A caller reports that a receptacle circuit tripped twice after a portable heater was plugged in. A damp ceiling tile is visible nearby.
>
> **What we know:** A receptacle circuit tripped after a portable heater was connected. A damp ceiling tile is visible nearby. The cause is not established.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the approved electrical procedure and a qualified person.

Question 2 | Intake & scope

Ask AI what information is missing and have it write three short questions for the requester or technician.

> **Job situation:** A caller reports that a receptacle circuit tripped twice after a portable heater was plugged in. A damp ceiling tile is visible nearby.
>
> **What we know:** A receptacle circuit tripped after a portable heater was connected. A damp ceiling tile is visible nearby. The cause is not established.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the approved electrical procedure and a qualified person.

Question 3 | Source review

Give AI the approved note, manual section, or procedure. Ask it to find the two or three useful points and give you the page or section number.

> **Job situation:** The service center has the caller's note and panel identifier, but no load inventory or authorized electrical test record.
>
> **What we know:** A receptacle circuit tripped after a portable heater was connected. A damp ceiling tile is visible nearby. The cause is not established.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the approved electrical procedure and a qualified person.

Question 4 | Source review

AI made a guess about the cause. Ask: "What in the note supports that? What is missing? How should we word it if we are not sure?"

Job situation: The service center has the caller's note and panel identifier, but no load inventory or authorized electrical test record.

What we know: A receptacle circuit tripped after a portable heater was connected. A damp ceiling tile is visible nearby. The cause is not established.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the approved electrical procedure and a qualified person.

Question 5 | Draft & critique

Ask AI to turn the checked facts into a short email, work-order update, or shift note. Tell it who will read it and what facts to include.

Job situation: AI suggests resetting the breaker again to see whether the problem clears.

What we know: A receptacle circuit tripped after a portable heater was connected. A damp ceiling tile is visible nearby. The cause is not established.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the approved electrical procedure and a qualified person.

Question 6 | Draft & critique

Ask AI to sort the draft into three groups: what we saw, what we think, and what a qualified person still needs to approve.

Job situation: AI suggests resetting the breaker again to see whether the problem clears.

What we know: A receptacle circuit tripped after a portable heater was connected. A damp ceiling tile is visible nearby. The cause is not established.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the approved electrical procedure and a qualified person.

Question 7 | Handoff

Ask AI to make a short review checklist for the supervisor or qualified person who must check the draft before it is used.

Job situation: The electrical supervisor needs a safe escalation note that does not authorize panel work by an unqualified person.

What we know: A receptacle circuit tripped after a portable heater was connected. A damp ceiling tile is visible nearby. The cause is not established.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the approved electrical procedure and a qualified person.

Question 8 | Handoff

Ask AI to find and flag names, access details, security information, contract details, or other information that should not be shared.

Job situation: The electrical supervisor needs a safe escalation note that does not authorize panel work by an unqualified person.

What we know: A receptacle circuit tripped after a portable heater was connected. A damp ceiling tile is visible nearby. The cause is not established.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the approved electrical procedure and a qualified person.

5. Use AI for a leak handoff

Plumbing | Foundations | 8 questions

Practice using AI to organize a water incident while keeping the source uncertain.

Scenario context

a leak above an occupied tenant space

Evidence source: ILLUSTRATIVE CASE · CEILING LEAK

Questions and changing situations

Question 1 | Intake & scope

Ask AI to clean up this request or note into a draft work order. Have it list the location, problem, facts, missing information, and next check.

> **Job situation:** A tenant reports water near a ceiling light after heavy rain. The nearest restroom is one floor above and a roof drain is nearby.
>
> **What we know:** Water is near a ceiling light after rain. The nearest restroom is one floor above and a roof drain is nearby. No source is confirmed.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the responsible trade and a safe site inspection.

Question 2 | Intake & scope

Ask AI what information is missing and have it write three short questions for the requester or technician.

> **Job situation:** A tenant reports water near a ceiling light after heavy rain. The nearest restroom is one floor above and a roof drain is nearby.
>
> **What we know:** Water is near a ceiling light after rain. The nearest restroom is one floor above and a roof drain is nearby. No source is confirmed.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the responsible trade and a safe site inspection.

Question 3 | Source review

Give AI the approved note, manual section, or procedure. Ask it to find the two or three useful points and give you the page or section number.

> **Job situation:** The work-control desk has the room number, time, weather, and a photo, but no confirmed source and no moisture reading.
>
> **What we know:** Water is near a ceiling light after rain. The nearest restroom is one floor above and a roof drain is nearby. No source is confirmed.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the responsible trade and a safe site inspection.

Question 4 | Source review

AI made a guess about the cause. Ask: "What in the note supports that? What is missing? How should we word it if we are not sure?"

Job situation: The work-control desk has the room number, time, weather, and a photo, but no confirmed source and no moisture reading.

What we know: Water is near a ceiling light after rain. The nearest restroom is one floor above and a roof drain is nearby. No source is confirmed.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the responsible trade and a safe site inspection.

Question 5 | Draft & critique

Ask AI to turn the checked facts into a short email, work-order update, or shift note. Tell it who will read it and what facts to include.

Job situation: AI labels the problem "clogged roof drain" because the roof drain is the closest item in the note.

What we know: Water is near a ceiling light after rain. The nearest restroom is one floor above and a roof drain is nearby. No source is confirmed.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the responsible trade and a safe site inspection.

Question 6 | Draft & critique

Ask AI to sort the draft into three groups: what we saw, what we think, and what a qualified person still needs to approve.

Job situation: AI labels the problem "clogged roof drain" because the roof drain is the closest item in the note.

What we know: Water is near a ceiling light after rain. The nearest restroom is one floor above and a roof drain is nearby. No source is confirmed.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the responsible trade and a safe site inspection.

Question 7 | Handoff

Ask AI to make a short review checklist for the supervisor or qualified person who must check the draft before it is used.

Job situation: The next shift needs to know what was protected, what was documented, and which trade must investigate.

What we know: Water is near a ceiling light after rain. The nearest restroom is one floor above and a roof drain is nearby. No source is confirmed.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the responsible trade and a safe site inspection.

Question 8 | Handoff

Ask AI to find and flag names, access details, security information, contract details, or other information that should not be shared.

Job situation: The next shift needs to know what was protected, what was documented, and which trade must investigate.

What we know: Water is near a ceiling light after rain. The nearest restroom is one floor above and a roof drain is nearby. No source is confirmed.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the responsible trade and a safe site inspection.

6. Use AI around life-safety impairments

Fire / Life Safety | Safety-critical | 8 questions

Practice using AI for documentation and escalation without bypassing an impairment procedure.

Scenario context

an offline fire-alarm point in an occupied wing

Evidence source: ILLUSTRATIVE CASE · OFFLINE SMOKE DETECTOR POINT

Questions and changing situations

Question 1 | Intake & scope

Ask AI to clean up this request or note into a draft work order. Have it list the location, problem, facts, missing information, and next check.

> **Job situation:** The fire panel shows one smoke-detector point offline in an occupied wing after ceiling work took place nearby.
>
> **What we know:** One smoke-detector point is offline. A ceiling contractor worked nearby. The cause is unknown and the building is occupied.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the designated responsible person and approved impairment procedure.

Question 2 | Intake & scope

Ask AI what information is missing and have it write three short questions for the requester or technician.

> **Job situation:** The fire panel shows one smoke-detector point offline in an occupied wing after ceiling work took place nearby.
>
> **What we know:** One smoke-detector point is offline. A ceiling contractor worked nearby. The cause is unknown and the building is occupied.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the designated responsible person and approved impairment procedure.

Question 3 | Source review

Give AI the approved note, manual section, or procedure. Ask it to find the two or three useful points and give you the page or section number.

> **Job situation:** The facilities desk has the panel message, point label, time, and occupancy status, but no confirmed cause.
>
> **What we know:** One smoke-detector point is offline. A ceiling contractor worked nearby. The cause is unknown and the building is occupied.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the designated responsible person and approved impairment procedure.

Question 4 | Source review

AI made a guess about the cause. Ask: "What in the note supports that? What is missing? How should we word it if we are not sure?"

Job situation: The facilities desk has the panel message, point label, time, and occupancy status, but no confirmed cause.

What we know: One smoke-detector point is offline. A ceiling contractor worked nearby. The cause is unknown and the building is occupied.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the designated responsible person and approved impairment procedure.

Question 5 | Draft & critique

Ask AI to turn the checked facts into a short email, work-order update, or shift note. Tell it who will read it and what facts to include.

Job situation: AI recommends bypassing the point until the next scheduled inspection.

What we know: One smoke-detector point is offline. A ceiling contractor worked nearby. The cause is unknown and the building is occupied.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the designated responsible person and approved impairment procedure.

Question 6 | Draft & critique

Ask AI to sort the draft into three groups: what we saw, what we think, and what a qualified person still needs to approve.

Job situation: AI recommends bypassing the point until the next scheduled inspection.

What we know: One smoke-detector point is offline. A ceiling contractor worked nearby. The cause is unknown and the building is occupied.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the designated responsible person and approved impairment procedure.

Question 7 | Handoff

Ask AI to make a short review checklist for the supervisor or qualified person who must check the draft before it is used.

Job situation: The responsible person needs a concise impairment escalation with no unnecessary security layout information.

What we know: One smoke-detector point is offline. A ceiling contractor worked nearby. The cause is unknown and the building is occupied.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the designated responsible person and approved impairment procedure.

Question 8 | Handoff

Ask AI to find and flag names, access details, security information, contract details, or other information that should not be shared.

Job situation: The responsible person needs a concise impairment escalation with no unnecessary security layout information.

What we know: One smoke-detector point is offline. A ceiling contractor worked nearby. The cause is unknown and the building is occupied.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: the designated responsible person and approved impairment procedure.

7. Use AI to review roof notes

Building Envelope | Intermediate | 8 questions

Practice turning inspection notes into a cautious summary without claiming a hidden water path is proven.

Scenario context

a roof-leak inspection summary

Evidence source: ILLUSTRATIVE NOTES · ROOF ZONE C

Questions and changing situations

Question 1 | Intake & scope

Ask AI to clean up this request or note into a draft work order. Have it list the location, problem, facts, missing information, and next check.

> **Job situation:** A ceiling stain appears 12 feet south of a patched roof penetration after a west-wind storm. No moisture reading was taken.
>
> **What we know:** An interior stain is south of a patched penetration. No moisture reading was taken. Water may travel along assemblies and roof access has fall-protection requirements.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a controlled inspection under the site access procedure.

Question 2 | Intake & scope

Ask AI what information is missing and have it write three short questions for the requester or technician.

> **Job situation:** A ceiling stain appears 12 feet south of a patched roof penetration after a west-wind storm. No moisture reading was taken.
>
> **What we know:** An interior stain is south of a patched penetration. No moisture reading was taken. Water may travel along assemblies and roof access has fall-protection requirements.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a controlled inspection under the site access procedure.

Question 3 | Source review

Give AI the approved note, manual section, or procedure. Ask it to find the two or three useful points and give you the page or section number.

> **Job situation:** The inspector has photos, weather notes, a roof-zone label, and a fall-protection requirement for roof access.
>
> **What we know:** An interior stain is south of a patched penetration. No moisture reading was taken. Water may travel along assemblies and roof access has fall-protection requirements.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a controlled inspection under the site access procedure.

Question 4 | Source review

AI made a guess about the cause. Ask: "What in the note supports that? What is missing? How should we word it if we are not sure?"

Job situation: The inspector has photos, weather notes, a roof-zone label, and a fall-protection requirement for roof access.

What we know: An interior stain is south of a patched penetration. No moisture reading was taken. Water may travel along assemblies and roof access has fall-protection requirements.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a controlled inspection under the site access procedure.

Question 5 | Draft & critique

Ask AI to turn the checked facts into a short email, work-order update, or shift note. Tell it who will read it and what facts to include.

Job situation: AI states that the patched penetration is definitely the water-entry point.

What we know: An interior stain is south of a patched penetration. No moisture reading was taken. Water may travel along assemblies and roof access has fall-protection requirements.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a controlled inspection under the site access procedure.

Question 6 | Draft & critique

Ask AI to sort the draft into three groups: what we saw, what we think, and what a qualified person still needs to approve.

Job situation: AI states that the patched penetration is definitely the water-entry point.

What we know: An interior stain is south of a patched penetration. No moisture reading was taken. Water may travel along assemblies and roof access has fall-protection requirements.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a controlled inspection under the site access procedure.

Question 7 | Handoff

Ask AI to make a short review checklist for the supervisor or qualified person who must check the draft before it is used.

Job situation: The building manager needs a handoff that distinguishes observations from the next safe inspection step.

What we know: An interior stain is south of a patched penetration. No moisture reading was taken. Water may travel along assemblies and roof access has fall-protection requirements.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a controlled inspection under the site access procedure.

Question 8 | Handoff

Ask AI to find and flag names, access details, security information, contract details, or other information that should not be shared.

Job situation: The building manager needs a handoff that distinguishes observations from the next safe inspection step.

What we know: An interior stain is south of a patched penetration. No moisture reading was taken. Water may travel along assemblies and roof access has fall-protection requirements.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a controlled inspection under the site access procedure.

8. Use AI with an IAQ complaint

Custodial / IAQ | Foundations | 8 questions

Practice organizing an occupant complaint without making unsupported health claims.

Scenario context

an indoor-air-quality complaint

Evidence source: ILLUSTRATIVE CASE · STALE-AIR COMPLAINT

Questions and changing situations

Question 1 | Intake & scope

Ask AI to clean up this request or note into a draft work order. Have it list the location, problem, facts, missing information, and next check.

> **Job situation:** Occupants say a training room smells stale in the afternoon. A wall sensor reads 1,250 ppm CO_2, but its accuracy is unverified.
>
> **What we know:** A room smells stale in the afternoon. CO_2 is 1,250 ppm from an unverified sensor. Occupancy was 18. Filter date and outdoor-air measurement are unknown.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: sensor verification, operating context, and the approved IAQ process.

Question 2 | Intake & scope

Ask AI what information is missing and have it write three short questions for the requester or technician.

> **Job situation:** Occupants say a training room smells stale in the afternoon. A wall sensor reads 1,250 ppm CO_2, but its accuracy is unverified.
>
> **What we know:** A room smells stale in the afternoon. CO_2 is 1,250 ppm from an unverified sensor. Occupancy was 18. Filter date and outdoor-air measurement are unknown.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: sensor verification, operating context, and the approved IAQ process.

Question 3 | Source review

Give AI the approved note, manual section, or procedure. Ask it to find the two or three useful points and give you the page or section number.

> **Job situation:** The coordinator has occupancy, time of day, filter date, and a complaint note, but no outdoor-air measurement or health assessment.
>
> **What we know:** A room smells stale in the afternoon. CO_2 is 1,250 ppm from an unverified sensor. Occupancy was 18. Filter date and outdoor-air measurement are unknown.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: sensor verification, operating context, and the approved IAQ process.

Question 4 | Source review

AI made a guess about the cause. Ask: "What in the note supports that? What is missing? How should we word it if we are not sure?"

Job situation: The coordinator has occupancy, time of day, filter date, and a complaint note, but no outdoor-air measurement or health assessment.

What we know: A room smells stale in the afternoon. CO_2 is 1,250 ppm from an unverified sensor. Occupancy was 18. Filter date and outdoor-air measurement are unknown.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: sensor verification, operating context, and the approved IAQ process.

Question 5 | Draft & critique

Ask AI to turn the checked facts into a short email, work-order update, or shift note. Tell it who will read it and what facts to include.

Job situation: AI drafts "the room is unsafe" from the single CO_2 reading.

What we know: A room smells stale in the afternoon. CO_2 is 1,250 ppm from an unverified sensor. Occupancy was 18. Filter date and outdoor-air measurement are unknown.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: sensor verification, operating context, and the approved IAQ process.

Question 6 | Draft & critique

Ask AI to sort the draft into three groups: what we saw, what we think, and what a qualified person still needs to approve.

Job situation: AI drafts "the room is unsafe" from the single CO_2 reading.

What we know: A room smells stale in the afternoon. CO_2 is 1,250 ppm from an unverified sensor. Occupancy was 18. Filter date and outdoor-air measurement are unknown.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: sensor verification, operating context, and the approved IAQ process.

Question 7 | Handoff

Ask AI to make a short review checklist for the supervisor or qualified person who must check the draft before it is used.

Job situation: The custodial or IAQ lead needs a respectful handoff that records the complaint without making a health diagnosis.

What we know: A room smells stale in the afternoon. CO_2 is 1,250 ppm from an unverified sensor. Occupancy was 18. Filter date and outdoor-air measurement are unknown.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: sensor verification, operating context, and the approved IAQ process.

Question 8 | Handoff

Ask AI to find and flag names, access details, security information, contract details, or other information that should not be shared.

Job situation: The custodial or IAQ lead needs a respectful handoff that records the complaint without making a health diagnosis.

What we know: A room smells stale in the afternoon. CO_2 is 1,250 ppm from an unverified sensor. Occupancy was 18. Filter date and outdoor-air measurement are unknown.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: sensor verification, operating context, and the approved IAQ process.

9. Use AI with irrigation data

Grounds / Irrigation | Foundations | 8 questions

Practice distinguishing a controller command from actual water delivery and coverage.

Scenario context

a dry landscape zone with a successful controller log

Evidence source: ILLUSTRATIVE CASE · IRRIGATION ZONE 7

Questions and changing situations

Question 1 | Intake & scope

Ask AI to clean up this request or note into a draft work order. Have it list the location, problem, facts, missing information, and next check.

> **Job situation:** The irrigation controller logged 20 minutes for Zone 7, but turf along the south walk is dry and the valve box is full of water.
>
> **What we know:** The controller logged 20 minutes of runtime. Turf is dry, a valve box is wet, and no flow or head-coverage reading is attached.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a safe field check of valve, flow, and coverage.

Question 2 | Intake & scope

Ask AI what information is missing and have it write three short questions for the requester or technician.

> **Job situation:** The irrigation controller logged 20 minutes for Zone 7, but turf along the south walk is dry and the valve box is full of water.
>
> **What we know:** The controller logged 20 minutes of runtime. Turf is dry, a valve box is wet, and no flow or head-coverage reading is attached.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a safe field check of valve, flow, and coverage.

Question 3 | Source review

Give AI the approved note, manual section, or procedure. Ask it to find the two or three useful points and give you the page or section number.

> **Job situation:** The grounds team has the controller log, zone map, weather, and a note that no flow or head-coverage reading is attached.
>
> **What we know:** The controller logged 20 minutes of runtime. Turf is dry, a valve box is wet, and no flow or head-coverage reading is attached.
>
> **What to check next:** The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a safe field check of valve, flow, and coverage.

Question 4 | Source review

AI made a guess about the cause. Ask: "What in the note supports that? What is missing? How should we word it if we are not sure?"

Job situation: The grounds team has the controller log, zone map, weather, and a note that no flow or head-coverage reading is attached.

What we know: The controller logged 20 minutes of runtime. Turf is dry, a valve box is wet, and no flow or head-coverage reading is attached.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a safe field check of valve, flow, and coverage.

Question 5 | Draft & critique

Ask AI to turn the checked facts into a short email, work-order update, or shift note. Tell it who will read it and what facts to include.

Job situation: AI recommends doubling the runtime and concludes that the controller is broken.

What we know: The controller logged 20 minutes of runtime. Turf is dry, a valve box is wet, and no flow or head-coverage reading is attached.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a safe field check of valve, flow, and coverage.

Question 6 | Draft & critique

Ask AI to sort the draft into three groups: what we saw, what we think, and what a qualified person still needs to approve.

Job situation: AI recommends doubling the runtime and concludes that the controller is broken.

What we know: The controller logged 20 minutes of runtime. Turf is dry, a valve box is wet, and no flow or head-coverage reading is attached.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a safe field check of valve, flow, and coverage.

Question 7 | Handoff

Ask AI to make a short review checklist for the supervisor or qualified person who must check the draft before it is used.

Job situation: The next grounds shift needs a short plan to check delivery, coverage, and water use without wasting more water.

What we know: The controller logged 20 minutes of runtime. Turf is dry, a valve box is wet, and no flow or head-coverage reading is attached.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a safe field check of valve, flow, and coverage.

Question 8 | Handoff

Ask AI to find and flag names, access details, security information, contract details, or other information that should not be shared.

Job situation: The next grounds shift needs a short plan to check delivery, coverage, and water use without wasting more water.

What we know: The controller logged 20 minutes of runtime. Turf is dry, a valve box is wet, and no flow or head-coverage reading is attached.

What to check next: The learner should use AI only for preparation and documentation, identify what is missing, and keep a human reviewer responsible for: a safe field check of valve, flow, and coverage.

Common Learner Output

- A practical work-order draft, email, summary, checklist, or handoff.
- A clear distinction between what was observed, what AI suggested, and what remains unknown.
- A named human reviewer and a next check before the output is used.
- Sensitive information removed or protected before sharing.
- A response written in the learner's own words and graded against the same five criteria.